Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 24)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe how sulfur can be removed from coal **before** combustion.

[2 marks]

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0 1 . 2 Describe the process of dry flue gas desulfurisation and explain how the product can be used.

[3 marks]

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0 1 . 3 In wet flue gas desulfurisation the flue gases are bubbled through sodium sulfite solution. Explain **one** advantage of this method over dry FGD.

[2 marks]

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0 1 . 4 Explain how each of the following reduces NOx emissions.

[3 marks]

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Fluidised bed combustion .....

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Catalytic converters .....

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Urea sprays .....

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10

0 2 Figure 1 compares the energy flows in a conventional gas-fired power station and a combined heat and power (CHP) station, each supplied with 100 units of fuel energy.

Figure 1

	Conventional power station	**CHP station**
Electricity output / units	40	35
Useful heat supplied to buildings / units	0	45
Waste heat lost / units	60	20

0 2 . 1 Calculate the overall efficiency of each station.

[2 marks]

Answer ..... %

0 2 . 2 Explain why the electrical efficiency of a CHP station is often deliberately kept below 40%.

[2 marks]

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0 2 . 3 Explain why CHP stations must be located close to the users of the heat.

[1 mark]

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0 2 . 4 Explain how each of the following features of an industrial heat exchanger increases its efficiency.

[3 marks]

Long narrow copper pipes .....

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Counter-current flow .....

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Reduced flow rate .....

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0 2 . 5 Explain why a single large spherical tank loses less heat than several small cubic tanks holding the same total volume of hot fluid.

[2 marks]

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0 3 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 3 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 3 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 4 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 4 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 4 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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0 5 Table 1 shows the efficiency and cost of four types of solar photovoltaic (PV) cell.

Table 1

PV cell type	Efficiency / %	Relative cost per m^2
Amorphous silicon (thin film)	8	1
Polycrystalline silicon	16	2
Monocrystalline silicon with anti-reflective surface	22	3
Multi-junction cell	40	25

0 5 . 1 Describe how a photovoltaic cell generates electricity.

[2 marks]

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0 5 . 2 Explain how anti-reflective surfaces and multi-junction cells increase efficiency.

[3 marks]

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0 5 . 3 Use Table 1 to suggest which cell type would be chosen for a large solar farm on cheap desert land and which for a satellite. Justify your choices.

[3 marks]

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0 5 . 4 Explain why solar PV output is intermittent **and** unpredictable.

[2 marks]

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0 6 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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0 6 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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0 6 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

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0 6 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

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0 6 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

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0 7 . 1 Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique.

[6 marks]

Table 1

Organisms	Technique	One limitation
Invertebrates active on the ground surface at night		
Invertebrates living in leaf litter		
Invertebrates on tree branches		

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0 7 . 2 Explain why a light trap cannot be used to prove that a moth species is absent from a site.

[2 marks]

0 7 . 3 Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

[2 marks]

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0 8 Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg^{-1}	Charge-discharge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

0 8 . 1 Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

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0 8 . 2 Give **two** other factors that affect the viability of a rechargeable battery.

[2 marks]

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0 8 . 3 Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

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0 8 . 4 Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

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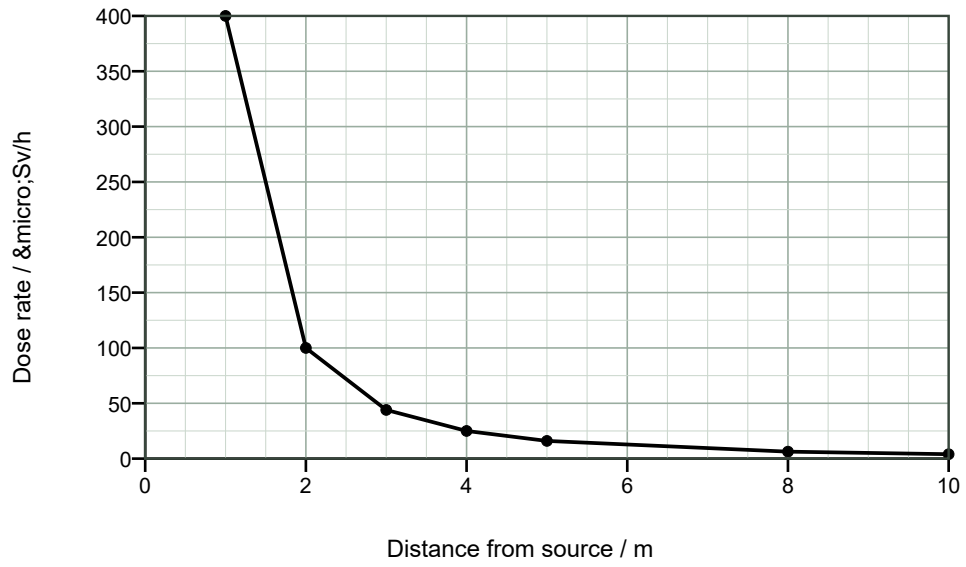
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0 9 Figure 1 shows the radiation dose rate measured at different distances from a small gamma source.

Figure 1



0 9 . 1 Use Figure 1 to show that the dose rate follows the inverse square law.

[2 marks]

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0 9 . 2 Calculate the dose rate at 20 m from the source.

[1 mark]

Answer µSv/h

0 9 . 3 Explain how each of the following reduces the exposure of workers handling radioactive sources.

[4 marks]

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Closed (sealed) sources .....

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Absorbing materials .....

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Reduced exposure time .....

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Worker monitoring .....

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0 9 . 4 Explain why potassium iodide tablets are issued to people near a nuclear accident.

[2 marks]

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0 9 . 5 State the meaning of ALARA.

[1 mark]

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1 0 . 1 As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

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1 0 . 2 State what is meant by the **stock** (resource base) of a mineral.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1 1 . 2 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS